

Study Design

Agentic Chart Review for Pediatric Asthma Guideline Adherence: Study Design

1. Study Overview

A multi-site retrospective observational study that develops, validates, and applies a question-driven agentic chart review framework to measure PCP adherence to NAEPP pediatric asthma guidelines. The framework is validated on two development sites (INPC, WCM) and applied across four sites to descriptively compare guideline adherence patterns, including a site with an established paper-based Easy Breathing program (Connecticut Children's) versus sites without (INPC, WCM, Nemours Children's).

Publication Plan

- **Paper 1 (Methods-primary):** Develop, validate, and apply the agentic asthma guideline adherence extraction framework. Descriptive 4-site adherence comparison (INPC, WCM, CT Children's, Nemours) as the primary use case, with a research question on whether data source type (HIE vs. single-system EHR) affects measured adherence and attribution patterns.
- **Paper 2 (Clinical, future):** Full effectiveness evaluation of paper Easy Breathing using difference-in-differences with propensity-matched cohorts. CT Children's (intervention) vs. WCM and Nemours (controls).

2. Sites and Data Sources

Development Sites

Site	Data Source	Note System	Role
Indiana (INPC)	Indiana Health Information Exchange	Multi-system HIE aggregation	Development + validation
Weill Cornell Medicine (WCM)	Epic EHR	Single-system (Epic)	Development + validation; Paper 2 control (no EB)

Application Sites

Site	Data Source	Note System	Condition
Connecticut Children's	EHR	Epic (assumed)	Intervention (paper Easy Breathing implemented)
Nemours Children's	EHR	Epic (assumed)	Control (no Easy Breathing)

Analytic Roles by Paper

- **Paper 1 (4-site descriptive comparison):** All sites contribute. INPC included with caveats — HIE data aggregates records from multiple systems, which may result in incomplete capture of within-visit documentation (e.g., missing action plans, education notes that exist in source EHRs but are not transmitted to the HIE). Differences in measured adherence at INPC may reflect data completeness rather than practice variation.
- **Paper 2 (effectiveness evaluation):** CT Children's (intervention) vs. WCM + Nemours (controls). INPC excluded due to HIE data incompleteness.

Validation Strategy

Develop agent on INPC + WCM with full annotation. Validate with smaller annotated samples at each application site (~15–20 patients) to quantify cross-site performance before full deployment.

3. Study Population

Inclusion Criteria

- Pediatric patients (ages 2–17) with documented asthma diagnosis (ICD-10 J45.x)
- At least 2 asthma-related encounters within the study period
- Longitudinal clinical notes available (progress notes, encounter notes, referrals)

Exclusion Criteria

- Patients with <2 notes in the observation window
- Patients with primary diagnosis of cystic fibrosis, bronchiectasis, or other chronic lung disease that would confound asthma management assessment

Age-stratified guideline handling: NAEPP guidelines differ by age group (0–4, 5–11, 12+). The question framework will use age-appropriate answer schemas (e.g., step therapy definitions, ICS dosing thresholds vary by age group) rather than separate question sets. *Clinicians (Fedele, Gregory, Hollenbach) should confirm age-specific guideline mappings during question framework development.*

(Criteria to be refined with clinical collaborators)

4. Unit of Analysis

Per-Patient-Period Design

The unit of analysis is a **patient-period**: a fixed observation window (e.g., 6 months) for each patient. A single patient may contribute multiple periods.

The agent extracts information at the **per-encounter level** within each period. These encounter-level extractions are then rolled up into period-level adherence summaries.

Example: Patient A has 4 asthma visits in a 6-month period. The agent reviews notes for each visit and determines:

- Asthma control assessed: 3/4 visits (75%)
- Step-appropriate therapy: 4/4 visits (100%)
- Asthma action plan documented: 1/4 visits (25%)
- Period-level composite adherence: derived from component rates

Paper 1: Common Observation Window

All eligible patient-periods across 4 sites during a shared calendar window (e.g., 2023–2025). No index date or intervention event — the analysis is cross-sectional by site, measuring baseline adherence patterns. Patients contribute all qualifying periods within the window.

Paper 2: Two Analytic Frames (Effectiveness Evaluation)

1. **Intention-to-treat (ITT):** First observation period per patient after the index date (paper EB implementation date at Connecticut Children's; matched calendar date at WCM and Nemours). All eligible patients included regardless of whether they received EB-guided care.
2. **Per-protocol:** All periods for patients with confirmed EB-guided encounters (Connecticut Children's) vs. matched periods at control sites (WCM, Nemours). Captures adherence trajectory over time and accounts for gradual EB adoption.

5. Question Framework

NAEPP guideline-derived questions organized in a tiered dependency structure, analogous to the 18-question lung cancer framework. Questions span seven domains:

Domain Structure (Preliminary)

Domain	Description	Example Questions
A. Asthma Control Assessment	Was impairment and risk assessed?	Daytime symptoms documented? Nighttime awakenings? SABA use frequency? Activity limitation? Exacerbation history?
B. Severity / Step Classification	Was severity or control level determined?	Severity classification documented? Current step level? Step-up or step-down indicated?
C. Pharmacotherapy	Was therapy concordant with step?	Controller medication prescribed? Consistent with assigned step? SABA prescribed as rescue? ICS-formoterol for SMART if step 3+?
D. Asthma Action Plan	Was written plan provided?	Action plan documented? Personalized to patient? Green/yellow/red zones specified?
E. Follow-up	Was appropriate follow-up scheduled?	Follow-up interval appropriate for control level? Referral to specialist if indicated?
F. Education	Was self-management education documented?	Inhaler technique reviewed? Trigger avoidance discussed? Self-monitoring instruction?
G. Comorbidity / Risk Factor Assessment	Were comorbid conditions affecting asthma assessed?	Allergic rhinitis evaluated? GERD assessed? Obesity

Domain	Description	Example Questions
		documented? Environmental exposures (tobacco smoke, allergens) reviewed?

Tiered Dependencies

Tier 0: Asthma diagnosis confirmed → Severity/control level

Tier 1: Control assessment components (A)

Tier 2: Step classification and therapy assessment (B, C)

Tier 3: Action plan, education, follow-up, comorbidities (D, E, F, G)

Tier 4: Concordance judgment (composite)

Specific questions, answer formats, and tier assignments to be finalized with clinical collaborators (Fedele, Gregory, Hollenbach).

Open question for clinicians: How should tier dependencies work across encounters within a patient-period? For example, if severity/step is classified at encounter 1, should the agent carry that forward and skip Tier 0–1 for subsequent encounters in the same period, or should each encounter be evaluated independently? Some visits may be follow-ups where severity is already established and the focus is control reassessment — the tier logic needs to account for this.

6. Development and Validation

Phase 1: INPC + WCM (Development)

1. **Cohort identification:** Pediatric asthma patients (ICD-10 J45.x) with longitudinal notes
2. **Sampling:** >30 patients per site (sample size TBD after question framework finalization — a larger question set across 7 domains will likely require more patients than the lung cancer study's 30/site to achieve stable per-question performance estimates), stratified by:
 - Asthma severity (intermittent, mild persistent, moderate persistent, severe persistent)
 - Age group (2–5, 6–11, 12–17)
 - Note volume (low, medium, high)
3. **Annotation:**
 - Two independent human annotators per patient-question instance

- Structured label schema: binary, categorical, or extracted value (matching lung cancer approach)
 - Discrepancy resolution via structured adjudication
 - Inter-annotator agreement reported (Cohen's kappa)
4. **Agent evaluation:**
 - Per-question accuracy and F1 (mirroring lung cancer Table 1 format)
 - Error analysis: locate errors vs. reason errors (mirroring lung cancer Figure 4)
 - Token cost analysis per patient-question
 5. **Cross-site generalization:** Compare performance on INPC (HIE, heterogeneous) vs. WCM (Epic, single-system) to assess robustness across data environments

Phase 2: Application Sites (Validation + Deployment)

1. **Site-specific validation:** ~15–20 annotated patients per application site (Connecticut Children's, Nemours)
2. **Performance calibration:** Report accuracy/F1 at application sites vs. development sites
3. **Full deployment:** Apply validated agent to complete eligible cohorts at both application sites

7. Downstream Analysis

Paper 1: Methods Paper (4-Site Descriptive Comparison)

Tier 1 analysis — descriptive, no causal modeling:

1. **Agent performance:** Per-question accuracy and F1 across development sites (INPC, WCM); cross-site generalization to application sites (CT Children's, Nemours)
2. **Adherence rates by site:** Per-domain and composite adherence rates across all 4 sites with 95% confidence intervals
3. **Data source effects:** Compare adherence and attribution patterns between HIE-based (INPC) and single-system EHR sites (WCM, CT Children's, Nemours) — does data source type systematically affect measured adherence or shift attribution toward DOCUMENTATION_GAP?
4. **Stratification:** By asthma severity, age group, insurance type, observation period
5. **Attribution analysis:** Distribution of concordance verdicts and attribution categories across sites (mirroring lung cancer Table 3)
6. **Temporal patterns:** Adherence trends across sequential periods within the common observation window

Presentation: Tables and forest plots. No regression models or causal claims.

Paper 2: Clinical Paper (Future — Effectiveness Evaluation)

Tier 2 analysis — causal inference:

1. **Propensity score matching:** Match patients across sites on demographics, asthma severity, insurance, baseline healthcare utilization
2. **Difference-in-differences:** Compare pre-to-post change in adherence at Connecticut Children's (around EB implementation date) against same time-window change at matched WCM and Nemours patients
3. **Sensitivity analyses:** Varying observation window length, matching calipers, parallel trends assumption testing

System Design

Agentic Clinical Chart Review (ACCR) Framework: System Design

1. System Overview

A general-purpose clinical reasoning agent modeled on the Claude Code architecture — a single orchestrator agent with tool access, subagent spawning, and pluggable "skills" (guideline-specific question frameworks). The system is EHR-note-focused but guideline-flexible: the data source is always longitudinal clinical records, but the guideline ingestion and question generation adapts to any disease area.

Two Operational Modes

1. **Guideline Ingestion Mode** — ingest a clinical guideline document, generate a question framework, refine through active learning with clinician feedback
2. **Chart Review Mode** — given a patient and question framework, extract structured answers with evidence provenance, then evaluate concordance

Deployment Path

The system starts as a research tool (retrospective cohort analysis, accuracy over cost) with a designed path to production deployment via model distillation and fine-tuning. Every chart review session produces structured reasoning traces as first-class output, enabling a training data flywheel: frontier model → traces → preference judgments → fine-tuned smaller models → self-hosted deployment.

Core Design Principles

- **Tool-use agent, not pipeline** — the orchestrator decides what to do next based on reasoning, not a fixed sequence
- **Capability-based subagents** — retrieval, extraction, verification are separate spawnable capabilities, not question-bound
- **Every session is training data** — tool calls, evidence retrieved, reasoning steps, final answers, and confidence scores are logged in structured format
- **Dual-track concordance** — deterministic rules for clear-cut logic, LLM-as-judge for nuanced clinical reasoning
- **Unified data abstraction** — notes and structured data accessed through the same tool interface regardless of backend (flat files, FHIR, OMOP, database)
- **Guideline skills as portable artifacts** — a new clinical domain means creating a new skill, not rebuilding the system

2. Guideline Ingestion Mode

Input

Clinical guideline document(s) — PDF, HTML, or markdown. Examples: NAEPP EPR-4, NCCN lung cancer guidelines, ADA diabetes standards of care.

Stage 1: Guideline Parsing & Question Generation

The orchestrator agent ingests the guideline with these tools:

Tool	Function
<code>read_guideline</code>	Parse guideline document, extract recommendations, decision trees, grading levels
<code>query_knowledge_base</code>	Look up related concepts — ICD codes, drug classes, lab tests — to ground questions in extractable clinical data
<code>generate_question_set</code>	Propose candidate questions with: question text, answer schema (binary/categorical/value/date), expected data sources (notes vs. structured), tier assignment, dependencies

Output: A candidate question framework — structured YAML/JSON defining each question, its answer type, dependencies, and which data sources are likely informative.

Stage 2: Active Learning Refinement

Iterative loop:

repeat:

1. Run agent on N sample patients (start with 5-10)
2. Score each question on:
 - extractability (did the agent find evidence?)

- confidence distribution (calibrated? or always 50/50?)
- inter-patient variance (does the question discriminate?)
- agreement with structured data (when both sources exist)

3. Surface to clinician:

- low-extractability questions → "Should we drop or rephrase?"
- low-confidence questions → "Is this genuinely ambiguous in notes, or is the question poorly scoped?"
- high-disagreement with structured data → "Which source is ground truth?"

4. Clinician edits: add/remove/split/merge/rephrase questions, adjust tiers

5. Update question framework

until: question set stabilizes (no changes in last round)

Stage 3: Annotation & Calibration

Once the question set is stable:

- Dual human annotation on development cohort (~30 patients/site)
- Inter-annotator agreement (Cohen's kappa) per question
- Questions with low kappa get revised or dropped — if humans can't agree, the agent shouldn't be evaluated on it
- Agent performance benchmarked against adjudicated gold standard

Skill Artifact

The question framework becomes a **skill** — a versioned, portable artifact that encapsulates:

- Questions with answer schemas
- Tier dependencies
- Retrieval hints (which note types, data sources are relevant per question)
- Concordance rules (deterministic + LLM judge prompts)
- Calibration benchmarks from development evaluation

Plugging in a new guideline means creating a new skill, not rebuilding the system.

3. Chart Review Mode — Agent Architecture

Orchestrator

A ReAct agent that receives a patient ID and a guideline skill. It:

1. Loads the question framework (tiers, dependencies, answer schemas)
2. Plans an execution strategy respecting tier dependencies
3. Spawns capability subagents as needed
4. Maintains a **patient context store** — accumulated facts extracted so far, shared across all subagent calls
5. Produces final structured output per question + overall concordance

Capability Subagents

Subagent	Role	Tools Available	Output
Retriever	Find relevant evidence for a set of information needs	<code>list_notes</code> , <code>search_notes</code> , <code>query_structured_data</code> , <code>query_knowledge_base</code>	Ranked evidence passages with source metadata (note ID, date, author, note type)
Extractor	Given evidence passages + question, produce structured answer with reasoning	(no external tools — works on provided context)	Answer value, confidence score, reasoning trace, cited passages
Verifier	Cross-check extracted answer against alternative data sources	<code>query_structured_data</code> , <code>read_note</code> (targeted)	Verification verdict (confirmed / contradicted / no data), discrepancy details

Execution Flow

Orchestrator loads skill, gets patient ID

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- |— Tier 0 questions (eligibility):
 - | |— Spawn Retriever → "find diagnosis confirmation, demographics"
 - | |— Spawn Extractor(s) for each T0 question on retrieved evidence
 - | |— Spawn Verifier → cross-check diagnosis against problem list/ICD codes
 - | |— Store confirmed facts in patient context
 - |
- |— Tier 1 questions (depends on T0 answers):
 - | |— Orchestrator checks dependencies — skip tier if T0 excludes patient
 - | |— Spawn Retriever → informed by T0 context
 - | | ("patient has moderate persistent asthma, find control assessment notes")
 - | |— Spawn Extractors for T1 questions
 - | |— Spawn Verifier where structured data exists
 - | |— Update patient context
 - |
- |— ... Tiers 2–N ...
- |
- |— Concordance evaluation (dual-track):
 - | |— Deterministic rule engine for clear-cut rules
 - | |— LLM-as-judge for nuanced concordance requiring clinical reasoning
 - | |— Per-rule verdict: CONCORDANT / NON_CONCORDANT / EXCLUDED
 - | |— Attribution category for non-concordant
 - | |— Disagreement between tracks → flagged for human review

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└ Output:

└ Per-question: {answer, confidence, evidence, reasoning_trace}

└ Per-rule: {verdict, attribution, supporting_questions}

└ Patient-level: composite adherence summary

└ Training log: full session trace (all tool calls, all reasoning)

Key Design Decisions

Why **capability-based**, not **question-based** subagents:

- A single Retriever call can gather evidence relevant to multiple questions in the same tier (e.g., one progress note may contain control assessment, medication changes, and follow-up plan)
- Avoids redundant retrieval — the lung cancer baseline agent re-read the same notes per question, driving up token cost
- Extractor subagents are lightweight (no tools, just reasoning over provided context) — cheap to parallelize

Why a **separate Verifier**:

- Structured data (med orders, labs, problem lists) is ground truth for certain questions — but not all
- Verifier only runs when structured data is available for that question type
- Catches the "documentation gap" problem surfaced in the lung cancer work (87.5% of non-concordance was documentation, not practice deviation)

Patient context store:

- Shared mutable state across tiers — each tier's extractions inform the next tier's retrieval
- Prevents the "satisfaction of search" error from lung cancer (agent stops too early because it doesn't know what's already been found)
- Implemented as a structured JSON object, not free text — so the orchestrator can reason over it programmatically

4. Unified Data Abstraction Layer

Tool Interface

The agent sees a single tool API regardless of whether the backend is INPC flat files, Epic FHIR, or a research database. Seven core tools:

Tool	Signature	Notes
<code>list_notes</code>	<code>(patient_id, note_type?, date_range?, author_role?)</code> → note metadata list	Filters without reading content. Cheap.
<code>read_note</code>	<code>(note_id)</code> → full text + metadata	Returns one note at a time to control context
<code>search_notes</code>	<code>(patient_id, query, date_range?)</code> → ranked passages with note IDs	Keyword or semantic search across all patient notes
<code>query_medications</code>	<code>(patient_id, date_range?, drug_class?)</code> → medication orders/administrations	Drug name, dose, route, frequency, start/stop dates
<code>query_labs</code>	<code>(patient_id, date_range?, test_name?)</code> → lab results	Test, value, units, reference range, date
<code>query_problems</code>	<code>(patient_id, active?)</code> → problem list entries	ICD codes, descriptions, onset dates, status
<code>query_encounters</code>	<code>(patient_id, date_range?, encounter_type?)</code> → encounter list	Date, type (office visit, ED, inpatient), provider, location

Backend Adapters

Tool API (what the agent sees)

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|— INPC Adapter: flat file corpus + structured tables

|— FHIR Adapter: DocumentReference, MedicationRequest, Observation,

| Condition, Encounter

|— Epic Clarity Adapter: direct SQL against Epic Clarity/Caboodle

|— OMOP CDM Adapter: note, drug_exposure, measurement,
condition_occurrence, visit_occurrence

Each adapter implements the same interface. Adding a new site means writing a new adapter, not changing the agent.

OMOP CDM as Natural Fit

If sites map to OMOP CDM:

- query_medications → drug_exposure table
- query_labs → measurement table
- query_problems → condition_occurrence table
- search_notes → note_nlp or full-text search on note table

A single OMOP adapter works across any OMOP-mapped site. The CodesetAgent work (separate paper) provides concept set authoring needed to define which drug concepts count as "ICS" or "SABA" per guideline.

Knowledge Base

A structured lookup service (not an LLM) for deterministic queries:

- Drug class memberships (e.g., "What medications count as step 3 controller therapy per NAEPP?")
- ICD/SNOMED code mappings
- Guideline-specific thresholds and definitions
- Step-therapy definitions

Implementation can range from simple lookup tables (MVP) to a formal knowledge graph (full version). The Retriever and Extractor subagents call this to ground reasoning in guideline-accurate facts rather than relying on parametric knowledge.

5. Concordance & Output Layer

Dual-Track Concordance Evaluation

Track	When to Use	How It Works
Deterministic rules	Clear-cut logic (was controller prescribed? does drug match step?)	Code over structured outputs. Fast, auditable, no hallucination risk.
LLM-as-judge	Nuanced concordance requiring clinical reasoning (e.g., "was the deviation clinically justified given this patient's comorbidities?")	LLM receives structured extracted answers (not raw notes) + guideline rule + patient context → verdict + reasoning

The two tracks complement each other:

- Deterministic handles the 70-80% of rules that are truly Boolean
- LLM judge handles cases where attribution requires reasoning (distinguishing **GUIDELINE_DEVIATION** from **PATIENT_FACTOR** often requires understanding why the clinician deviated)
- Both produce the same output schema (verdict, attribution, reasoning)
- **Disagreement between tracks** flags cases for human review — and becomes high-value RLHF training signal

Rule Definition Format

rule_id: NAEPP_C1

name: "Controller therapy concordant with step"

track: deterministic # or "llm_judge" or "dual"

inputs: [B1_step_classification, C1_controller_prescribed, C2_controller_matches_step]

logic: |

if B1 == EXCLUDED → EXCLUDED

if C1 == false → NON_CONCORDANT (attribution: MISSING_THERAPY)

if C2 == false → NON_CONCORDANT (attribution: STEP_MISMATCH)

else → CONCORDANT

Rules are defined as part of the guideline skill — versioned, testable, inspectable by clinicians.

Attribution Taxonomy

Standardized across guidelines, extensible per domain:

Category	Meaning
DOCUMENTATION_GAP	Relevant action may have occurred but isn't documented in available notes/data
GUIDELINE_DEVIATION	Documentation shows clinician made a different choice than guideline recommends
PATIENT_FACTOR	Documented clinical reason for deviation (allergy, contraindication, patient preference)
SYSTEM_FACTOR	Institutional/access barrier (formulary restriction, referral delay, insurance)
INSUFFICIENT_DATA	Not enough information to make a determination — distinct from documentation gap

Output Levels

1. **Question-level** — {question_id, answer, confidence, evidence, reasoning_trace}
2. **Rule-level** — {rule_id, verdict, attribution, supporting_questions, track_used}
3. **Summary-level** — composite adherence scores per domain and overall, with confidence intervals

Summary level aggregates for descriptive analysis (Paper 1). Rule-level feeds future computable quality measure definitions.

Confidence Calibration

- **Extraction confidence** — per question, from the extractor. Calibrated against development set (target: when agent says 90% confident, correct ~90% of the time)

- **Concordance confidence** — per rule, derived from extraction confidences of input questions. Low-confidence inputs flag the rule verdict for review
- Low-confidence outputs feed back into the active learning loop (RLHF training signal)

6. Training Data Flywheel

Trace Capture Format

Every chart review session automatically logs:

session:

patient_id: P123

skill: naepp-asthma-v2

model: claude-opus-4-6

steps:

- agent: orchestrator

action: plan_execution

reasoning: "Patient confirmed asthma dx, moderate persistent. Moving to Tier 1."

- agent: retriever

action: search_notes

params: {query: "asthma control symptoms", date_range: "2025-06-01/2025-12-01"}

result: [note_47, note_62, note_89]

- agent: extractor

question: A1_daytime_symptoms

evidence: ["...shortness of breath 2-3x/week..." (note_62, line 14)]

reasoning: "2-3x/week exceeds 'not well controlled' threshold of >2 days/week"

answer: {value: "not_well_controlled", confidence: 0.92}

- agent: verifier

question: C1_controller_prescribed

extracted: {value: "fluticasone 110mcg BID"}

structured: {drug_exposure: "fluticasone propionate 110mcg, BID, start 2025-07-15"}

verdict: confirmed

outputs:

questions: [{question_id, answer, confidence, evidence_ids, reasoning}]

concordance: [{rule_id, verdict, attribution}]

Training Stages

Stage 1: Knowledge Distillation (immediate)

Frontier model runs chart review → produces (input, reasoning trace, answer) tuples → SFT on open-weight base model (e.g., Llama 3.x, Qwen 2.5). Goal: 80% of frontier accuracy at 10% of the cost.

Stage 2: RLAIF (scales first)

Frontier model judges reasoning traces from the distilled model:

- "Did the retriever find the most relevant evidence?"
- "Is the reasoning clinically sound?"
- "Does the answer match the evidence?"

Generates preference pairs (better trace, worse trace) automatically. Train with DPO on these pairs. Runs continuously as patient reviews accumulate.

Stage 3: RLHF (high-value, targeted)

Clinicians review cases surfaced by the active learning loop:

- Low confidence extractions
- Verifier contradictions (extracted ≠ structured data)

- Concordance edge cases
- Dual-track disagreements

Gold preference pairs, more expensive but higher signal. Reserved for cases RLAIF can't resolve.

Stage 4: GRPO for reasoning (extension)

Sample multiple reasoning paths for the same extraction. Reward traces that reach correct answers. Particularly valuable for multi-step questions (e.g., "Is therapy step-appropriate?" requires: identify current step → identify prescribed meds → check alignment with guideline). Many valid reasoning paths; GRPO explores them.

Stage 5: LoRA adapters for retrieval (extension)

Fine-tune retrieval subagent specifically on relevance judgments — which notes/passages were actually used in correct extractions vs. retrieved but unused. Separate adapter from the reasoning model. Can be guideline-specific if retrieval patterns differ across clinical domains.

Model Progression

Phase 1: Frontier API (Claude/GPT) — full accuracy, high cost

↓ knowledge distillation (Stage 1)

Phase 2: Fine-tuned open-weight (70B) — near accuracy, moderate cost

↓ RLAIF + RLHF preference tuning (Stages 2-3)

Phase 3: Preference-tuned open-weight — improved reasoning, same cost

↓ GRPO + LoRA specialization (Stages 4-5)

Phase 4: Specialized models per capability

— retriever (8B + LoRA)

— extractor (70B tuned)

— verifier (8B)

↓ quantization + serving optimization

7. Component Roadmap

Each component has three implementation levels. Start at the level that matches current needs; upgrade when triggers are hit.

Component	Full Version	Simplified Version	MVP (Deferred)	Simplification Rationale
Orchestrator	ReAct agent with subagent spawning, patient context store	ReAct agent, no subagents, in-context state	—	Subagents add orchestration complexity; single agent proven in lung cancer work
Subagents	Retriever + Extractor + Verifier as independent capability agents	Retriever + Extractor combined; no Verifier	Single monolithic agent	Capability separation only pays off at scale; combined agent simpler to debug
Guideline Ingestion	Automated parsing → question generation → active learning loop	Semi-automated : LLM proposes, clinician edits, test on samples manually	Hand-crafted questions with clinicians (YAML file)	Only one guideline needed initially; automate when adding guideline #2
Active Learning	Formal extractability scoring, confidence calibration, systematic surfacing	Informal: run on samples, eyeball results, discuss with clinician	Skip — define questions once	Active learning most valuable when question set is large or guideline is unfamiliar
Note Tools	<code>list_notes</code> + <code>read_note</code> + <code>search_notes</code> (keyword + semantic)	<code>list_notes</code> + <code>read_note</code> + <code>search_notes</code> (keyword only)	<code>list_notes</code> + <code>read_note</code> only	Search is the biggest accuracy driver (lung cancer: 83.5%)

Component	Full Version	Simplified Version	MVP (Deferred)	Simplification Rationale
				→ 92.4%); worth including early
Structured Data Tools	query_medications, query_labs, query_problems, query_encounters	query_medications + query_problems only	None — notes only	Many asthma questions answerable from notes; structured data is accuracy boost, not requirement
Data Abstraction Layer	Unified interface with INPC, FHIR, OMOP, Clarity adapters	One adapter per active site, shared interface	Hardcoded INPC access	Abstraction only needed when site #2 comes online
OMOP Integration	Full OMOP CDM adapter with CodesetAgent concept sets	OMOP adapter with manual concept sets	None	Depends on whether sites have OMOP mappings available
Knowledge Base	Structured knowledge graph with drug classes, ICD mappings, guideline thresholds	Lookup tables (JSON/YAML) for drug classes and code mappings	None — agent uses parametric knowledge	Add if errors cluster on factual lookups (drug classifications, code mappings)
Concordance — Deterministic	Rule engine with versioned rule definitions per guideline skill	Hardcoded rules in Python	None — use LLM judge only	Deterministic rules are an optimization for speed and auditability
Concordance — LLM Judge	Fine-tuned judge model with reasoning traces,	Frontier model as judge (prompt-based, like lung cancer)	Same as simplified	Prompt-based judge already validated in lung cancer work

Component	Full Version	Simplified Version	MVP (Deferred)	Simplification Rationale
	calibrated confidence			
Dual-Track Concordance	Both tracks run, disagreements flagged for review + RLHF	Run one track only (LLM judge for nuanced rules, deterministic for Boolean)	LLM judge only	Dual-track mainly valuable for generating high-quality disagreement training data
Attribution Taxonomy	5-category standardized taxonomy with per-guideline extensions	3 categories: DOCUMENTATION_GAP, GUIDELINE_DEVIATION, INSUFFICIENT_DATA	Free-text attribution	Simplified taxonomy covers ~90% of cases
Confidence Calibration	Calibrated per-question and per-rule confidence, Platt scaling on dev set	Raw model confidence scores (uncalibrated)	None	Calibration requires sufficient dev set size; raw scores usable for ranking
Trace Logging	Structured YAML/JSON with full session traces, tool calls, reasoning	Structured JSON — answers + evidence only, no intermediate reasoning	Unstructured logs	Full traces essential for training flywheel; simplified still useful for evaluation
Knowledge Distillation	Frontier → SFT on open-weight model with curriculum learning	Frontier → basic SFT on traces	None — frontier only	Requires ~200+ patient traces to be worthwhile
RLAIF	Continuous preference pair generation +	Batch: accumulate	None	Requires distilled model

Component	Full Version	Simplified Version	MVP (Deferred)	Simplification Rationale
	DPO training loop	traces, periodic DPO runs		(Stage 1) to exist first
RLHF	Clinician preference judgments on active-learning-surfaced cases	Clinician review of random sample of errors	None	Most expensive; reserve for after RLAIIF plateau
GRPO	Multi-path sampling with outcome-based reward for reasoning	—	None	Extension — only after preference tuning shows reasoning gaps
LoRA Retrieval	Per-guideline retrieval adapters trained on relevance judgments	—	None	Extension — only if retrieval is the accuracy bottleneck
Multi-Site Deployment	Portable skill + adapter per site, cross-site performance monitoring	Manual replication with site-specific tweaks	Single site	Abstraction layer prerequisite

Starting Configuration

For Paper 1 (NAEPP asthma, INPC + WCM development):

- Orchestrator: **Full** (subagents provide the token cost reduction needed for multi-site)
- Subagents: **Simplified** (Retriever + Extractor combined; skip Verifier until structured data tools exist)
- Guideline Ingestion: **Simplified** (LLM proposes questions from NAEPP, clinicians refine)
- Active Learning: **Simplified** (run on samples, iterate with clinicians)
- Note Tools: **Full** (search is critical — lung cancer proved this)
- Structured Data Tools: **Simplified** (`query_medications` + `query_problems` — most impactful for asthma)
- Data Abstraction: **Simplified** (INPC adapter + WCM adapter, shared interface)

- Knowledge Base: **Simplified** (lookup tables for NAEPP drug classes and step definitions)
- Concordance: **Simplified** (LLM judge primary; deterministic rules for obvious Boolean checks)
- Trace Logging: **Full** (capture everything from day one — traces don't expire)
- Training Pipeline: **Deferred** (accumulate traces; train after sufficient data)